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NUMBER 8.

SATURDAY, AUGUST 22, 1840.

VOLUME I.



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THE HOWTH LIGHTHOUSE.

The bold and nearly insulated promontory called the Hill of Howth, which forms the north-eastern terminus of the Bay of Dublin, would in itself supply abundant materials for a topographical volume—and a most interesting work it might be made. For the geologist, botanist, and naturalist, it has an abundant store of attractions, while its various ancient monuments of every class and age, from the regal fortress, the sepulchral cairn, and the cromleac of Pagan times, to the early Christian oratory, the abbey and the baronial hall of later years, would supply an equally ample stock of materials for the antiquary and the historian. With all, or most of these features, we propose to make our readers somewhat familiar in our future numbers; but our present purpose is only to give some account of one of its most recently erected structures—the singularly picturesque and beautiful lighthouse, which we have attempted to depict in our prefixed illustration.

The Baily lighthouse, as it is popularly called, is situated at the eastern extremity of Howth, on a nearly perpendicular rock, whose vertex is elevated one hundred and ten feet above high-water mark. This rock, which is nearly insulated, is the terminus of a long and narrow peninsula of still higher altitude, which stretches out into the sea from the eastern end of the promontory, and whose cliffs are equally precipitous on both sides, so that the most striking and romantic views of the lighthouse can be had from various points, in some commanding the horizon-bound sea, and in others the Bay of Dublin, with all its delightful sceneries of wooded country and mountain ranges. The view which we have chosen for our illustration is taken from the northern side of this peninsula, that presented from the other side having been already published in several popular works; but we trust that this view will not be deemed less striking or picturesque; and we are of opinion that a more romantic subject of its kind is not to be found in the empire.

The lighthouse is itself an object of great interest and beauty, and is constructed according to the most approved models of modern times. Its form is that of a frustrated cone, supporting a lantern which exhibits a fixed bright light. The illumination, according to the system now generally adopted by the Trinity-house, is produced by a set of reflectors ground to the parabolic form, in the foci of which twenty large oil lamps are placed: an outer gallery, lightly but securely railed, surrounds the dome. Connected with the building on its east side, there is a large room, which opens by folding doors on a platform, and where an excellent telescope is kept, by means of which the shoals which obstruct the entrance to the bay may be distinctly observed—namely, the great Kish, and the Bennet and Burford banks, which are links of the chain extending along the Wicklow and Wexford coasts, and called the Irish grounds. These, though not visible, are distinctly marked in stormy weather by the surf, which breaks over them with uncommon violence, and form a dangerous obstruction to the approach to the bay.

The Baily lighthouse was erected by the Ballast Board of Dublin in 1814, previous to which time the Howth light, as it was commonly called, stood on a hill considerably more to the north, and at an elevation of more than three hundred feet above sea level. This circumstance of its great elevation, led, however, to its being abandoned, and the erection of the Baily lighthouse in its place, as it was found to be frequently involved in clouds and mist, while lower stations were clear and well defined.

The Baily lighthouse is a spot of no less antiquarian than picturesque interest. Its name, which is cognate with the Latin *ballium*, is derived from an ancient circular stone fortress which encircled the apex of the rock, and of which considerable remains existed previous to the erection of the

present buildings. This great keep was fortified by three earthen walls, with deep intervening ditches placed at the entrance to the narrow peninsula, and by extending from one side of it to the other, cut it off completely from the promontory. These works still remain, though in a very ruinous state; yet they are sufficiently distinct to mark their purpose, and to convey a good idea of the style of military defensive works in use in extremely remote times. They will be found marked on the Ordnance map.

In the popular traditions of Howth, these works—like most others in Ireland, the real origin of which has been forgotten—are ascribed to the Danes, a remnant of whom, after the battle of Clontarf in 1014, were supposed to have fortified themselves in this peninsula, till they were carried off in their vessels. But such tradition is wholly opposed to history, and the works themselves exhibit sufficient evidences of its fallacy; they belong to a much earlier age, being nothing less than the remains of Dun-Criomthan (pronounced Dun-Criffan), the fortress of Criomthan Nia-nair, who, according to our ancient histories, ascended the throne of Ireland in the year 74, and who, after being dethroned, died in this fastness in the year 90, after a reign of sixteen years. His sepulchral cairn—crowning the summit of Sliabh-Martin, the highest pinnacle of the ancient Bin-edair—is still to be seen.

A century or two more will wholly obliterate these remains of the once powerful prince and warrior Criomthan; but his celebrity belongs to history, and will not thus pass away. It was in the third year of his reign that Agricola fortified the bounds of the Roman empire in Britain from the incursions of the Picts and Irish, the latter, it is said, led by the monarch Criomthan himself, who, according to our annalist, returned to Ireland, loaded with spoil, as thus stated in the record of his death in the Annals of the Four Masters:—

“Criomthan Nia-nair, sixteen years monarch of Ireland, died, after his illustrious foreign expedition. It was from that expedition he brought home the noble spoils; the golden chariot, the golden chess-board studded with three hundred sparkling gems, and the *ceth-criomthan*, which was a parti-coloured shirt, interwoven with gold. He also brought with him a battle-giving sword, having various figures of serpents engraved upon it, and inlaid with gold; a shield embossed with bright silver; a spear which gave an incurable wound; a sling from which no erring cast could be thrown; two hounds linked together by a chain of silver; together with many other valuable rarities.”

How long after this period Dun-Criomthan existed as a fortress, it would perhaps be impossible now to ascertain, but from the following record in the Annals above quoted, it would appear to have been preserved at least for six centuries:—

“A. C. 646. The battle of Dun-Criomthan was gained by Conall and Kellach (co-monarchs of Ireland), the two sons of Maolcobha, over Aongus, the son of Donall. Aongus was killed in this battle, as was also Cathasach, the son of Donall, his brother.”

These notices, which have not hitherto appeared in an English form, of a highly interesting historical remain, not previously identified by the antiquarian topographer, will, it is hoped, impart a new interest to the Baily of Howth; but, independently of such claims on our attention, its singular picturesqueness should have made it long since not only more familiarly known to the visitors of our capital, but also to ourselves.

JOHNNY HALFACRE; OR, THE VALUE OF TIME.

BY MARTIN DOYLE.

Statesmen and professional men, whether occupying stations of eminence, or struggling to attain them, duly estimate the importance of time; they know the value of an hour too well to mis-spend it. The lawyer of high practice, during the term season, steadily pursuing his laborious studies, and determined to overcome every difficulty in his pursuit of professional rank and wealth, rises early, and borrows from the night so many of those hours which are spent in rest and sleep by men of less mental activity, that he leaves himself but a very contracted measure of time for those essential purposes. As to dining out with friends at this period of care and labour, he rarely ventures to indulge in such a recreation; or if he does on some very particular occasion, such is the discipline of his mind, such the strength of his self-denying habits, that he can rise from the table at a prescribed moment, and with a cool lawyer-like head apply to his nocturnal labours as if there had been no interruption of an exciting nature.

The physician—I do not mean him who is regularly called out of church, or from the social party, by his servant, under the pretence of a pressing call, but the real and laborious practitioner, to whom minutes are money and fame—will not idle away an hour; neither will the sober steady shopkeeper, until he has realized an independence, absent himself from his counter as long as there is a reasonable chance of a customer dropping in; nor the operative mechanic, who has to finish his piece of work within a prescribed time, and who will contrive to do it even in despite of all the petty interruptions to which he is liable.

Time is proportionably valuable to the meanest peasant who possesses a cabbage garden, and if properly estimated and applied, will add to his comforts in a degree of which, he who is habitually uncalculating and unthrifty in this respect can have but little notion.

This I am anxious to impress upon the class of labourers, many of whom I hope can read what I write, for in them I take an especial interest, probably because they are the least cared for of any class in the community. Some of them perhaps will say, with a show of reality, “If *our* time were to bring us in such profits as the counsellor and the doctor make, we would be busy too, and no one would see us standing idle, sitting on a ditch side, or smoking and coshering by the fireside, or talking to the neighbours, of a wet day, in a forge. If we could be coining guineas as easily as the likes of them makes the money, sitting in their soft chairs, and never doing a hand’s turn of work that would tire their limbs, we would; but what *could* we make, after our regular day’s work, if we can get that same, out of a bit of a garden, that would better us any thing to signify?”

Now, I shall show them by actual facts what they *could* in many cases do.

Johnny Halfacre is a little farmer, whom I occasionally see, and who, being in no way connected with me, nor even conscious that I am particularly observing him, goes on in his own way, without any hint or encouragement from me, or indeed from any one else, as far as I can perceive.

Johnny two years ago had not as much land as would correspond with his name, which is really genuine; he had for several previous years but a rood, including the site of his house, and a shed for a pig, and some poultry; but this rood produced more than half an acre usually does with many, and entirely by his good management and judicious application of *time*.

Johnny had exactly five shillings a-week, paid in full every Friday evening, from his employer, for Johnny never had time to be sick, far less to be drunk, and always avoided broken days, by contriving in-door work, at Mr B.'s, in wet weather; his wife, who had two children, washed occasionally for a neighbour's family, thus adding two shillings and sixpence each week to their income, and the contribution of additional suds to the dunghill; but in other respects they had no advantage over other labourers. Their own little garden added greatly to the support of the family, by judicious cropping and excellent management. Johnny had every year some drills of very early ash-leaved potatoes put down in January, if possible, which he either sold at a very high price in summer at a neighbouring town, or consumed as he found most economical; and his early sowing of potatoes was far better than the more common practice of the Irish cottier, who leaves his garden uncropped with them until March or April, with the view of obtaining a more abundant crop (but of inferior quality) at a late season, when they might be purchased at a mere trifle, and that, too, without the advantage of a second crop of any description to succeed them. Johnny had too much sense for this: he began to dig his dish of potatoes for dinner in the first or second week in July, when his neighbours were half starving, or paying exorbitantly for oatmeal and old potatoes; and as he dug out his crop, he either sowed turnips, with a little ashes and a sprinkling of dung, or planted borecole for the winter; generally he had some of both, for he found turnips good for his own table in winter, and profitable for the support of some poultry, of which I shall take notice soon. He had also every variety of common kitchen vegetables in small patches, continually changing places, and thus improving the soil; he had, besides, two hives of bees; and for the sake of the *straw*, as well as for rotation, and the support of his pig and poultry, a little rye, vetches, or clover.

Johnny, however, only worked in the garden in the evening, after his ordinary day's work, or, in summer, at sunrise; yet there never was a weed to be seen in it, for they never had time to grow: by using the hoe for a few moments now and then, they were always kept down, and every waste blade and briar and useless sod around the hedge which enclosed it, was carefully pared and burnt for manure.

He had worked in the large garden of a gentleman who kept an English gardener, who had taught Johnny the use of a *sprong* in preference to a spade for turning up the earth, especially when too hard for the latter implement; and though the handle was short, and, according to my own notion, fatiguing to the back, the fact was, that Johnny soon preferred it for dispatch and correctness of operation to the long-handled spade which all my other neighbours use. When he cut his own rye or other corn, the ground was usually so hard that a broad spade could not enter it: but Johnny quickly turned it up and broke it with his sprong, and then completely pulverized it with what the Englishman called a beck, a three-forked hoe, which, acting like the long tines of a harrow, loosened and rendered the whole perfectly fine, while it brought any latent roots of couch (or scutch grass) that might have escaped on former occasions, to the surface.

Johnny's various vegetables greatly assisted his housekeeping. He had often a good bowl of soup, flavoured with leeks, onions, carrots, &c., made with the least conceivable portion of meat, but thickened with barley, properly shelled, and prepared like French barley, but at only one-third of the price of that which is sold under such denomination in the shops; and his family always breakfasted on porridge, or coarse bread of their *own baking*, with or without milk, according to

circumstances—for Johnny at this time had no cow—sometimes washed down with a cup of tea, and more generally in winter with a mug of light and good table beer, which the Englishman taught Johnny to brew at Mr B.'s brew-house. Half a bushel of malt, with a quarter of a pound of hops, produced ten gallons of *unadulterated* beer which could not be bought any where, and the grains (given to his pig) fully counterbalanced the cost of fuel. Even at this time he killed a pig every year, and never wanted a small supply of salt meat for his cabbage or beans, which with this combination of flesh went farther in this way towards the actual supply of his dinner, and sometimes of his supper too (for any remainder of the dinner was heated and peppered up for the supper, with the addition of a broken loaf, or a skillet full of potatoes), than can be imagined by the poor man who has never cultivated his garden in the same manner—whose cabbages are of little value from want of bacon, and whose allotment, producing but one crop instead of two each year, is thus of but half its proper value to him; besides, with him potatoes succeed potatoes continually, until the ground becomes sick of yielding them.

But, further, Johnny Halfacre's garden, in which he seldom ceased from doing something in the summer evenings as long as daylight lasted, greatly aided in supporting his pig at that time when food is so dear and scarce for swine. The tops of blossoming bean-stalks (by the plucking off of which the crop is improved) and other vegetable waste, besides vetches and rye—the latter both in the green and ripe state—gave him sufficient food to keep the pig in fair order, with a little help from other sources; and the pig, by being always well littered, and supplied with this food, gave a return in most excellent manure, which with other sources of a similar kind, and the economical distribution of crops, supplied the entire garden with fertilizing matter.

What the other means of providing manure were, ought to be mentioned, for the man's system is of such easy application that it only requires to be stated in order to be followed.

For two or three evenings in the summer before last, I perceived Johnny Halfacre without his coat, rolling a wheelbarrow frequently from an adjacent common to a corner of his garden separated from the road by an old weather-beaten paling. When I had leisure to see what he had been doing at this time, I found that he had marked off an oblong space for four geese and a gander, which he had bought from Bridget Gozzard at rather a high price, partly for the sake of their powerful manure, which, combined with other substances, is good for stimulating the growth of vegetables, as well as for the profit which he expected to realize by rearing goslings for the market. Johnny was aware that fat green geese are worth from six to ten shillings each, in the very early season in the great English markets, and are also profitable if reared for the stubbles at Michaelmas; and he did not see why he and his industrious wife should not realise a profit as well as English housewives by the breeding of such poultry, when a steam-packet and a rail-road could take them off even to London in a few hours. Cocks and hens would ruin his own garden, and bring him into disputes with his neighbours—he had the advantage of a run on the common for geese—there was a pond of water near his house—and therefore he gave them and ducks the preference. He first built his back wall two feet and a half high and ten feet in length, with the sods from the common, and then put down ten upright stakes in front, every pair answering for the jambs of each compartment, with a board stretching the whole length across, and which formed the front support of his rustic roof; from this board he laid rafters to the top of the back wall, and having first interwoven some small branches of a tree through these rafters, he laid as many scraws (thinly pared grassy sods) as secured the whole roof from rain. The jambs were then contracted to a narrow opening, for the sake of shelter and warmth, by more sods laid one over the other.

By this simple process of construction he formed a separate chamber for each bird, with a yard in front six feet broad and ten long, and with an opening through the paling at the road side, by which

the inmates could go in and out at their pleasure. His rye assisted in feeding them, and he also cultivated grey peas for them, which are excellent for fattening; and with cabbage and lettuce leaves, the pods of beans, and other green food, he afterwards kept them in high condition; and in the succeeding year, when other young geese were dying of disease, occasioned by want of shelter, and from *starvation*, his were thriving.

And to the credit of this worthy man and his wife I must mention, that the feather-plucker was indignantly sent away from his door whenever he came round for the execrable purpose of plucking the geese alive. Johnny's wife would as soon have let him pull out the hairs of her own head, as give up one of her birds to his barbarous hands; and the consequence was, that while their neighbours' geese were miserably crawling about, with dragging and mutilated wings and smarting bodies, until many of them died, in their miseries invoking as it were in their dying screams shame and curses on their unfeeling owners, Johnny Halfacre's geese strutted about on the common, with an independent and unconstrained step, as if conscious of their security from the tortures to which their fellows had been doomed.

HOW JOHNNY HALFACRE BECAME A LITTLE FARMER.

If it be true, and it unquestionably is, that "he who despiseth small things, shall fall by little and little," the converse is, I think, no less so—that he who pays attention to little matters will rise by degrees.

Mr B. having narrowly observed Johnny's general good conduct and extreme industry as a common labourer, put him in possession, two years ago, of a field adjoining his cottage and garden, which contains about six statute acres, and which fortunately was in good condition.

Johnny at first was afraid to accept the tempting offer, at which any other labourer would have jumped, on the sincere and modest plea that he had no capital for such a weighty speculation. He did not wish to grasp at more than he could properly manage; but Mr B. set him at ease, by telling him that he considered health, industry, and skill, sufficient capital for Johnny to possess, as he himself would not only build a barn, cow-shed, ass-house, and pig-styes, but put the boundary fence into perfect order (according to the frequent practice of *British* landlords), and lend Johnny a sum sufficient for the purchase of every thing necessary to give him a good start, charging him only five per cent on the advances. Mr B., who in riding over his property often "went by the field of the slothful," which "was all grown over with thorns and nettles that covered the face of it, and the stone wall whereof was broken down," wished to render Johnny an exemplar of superior management to other tenants.

I shall not trouble the reader with all the details of Johnny's management during the two last years, but shall very briefly notice those particulars of husbandry which are new to my countrymen of the same class. He has not subdivided the field, nor does he intend to do so, as he values every foot of it too much for such waste. He does not keep a horse, nor will he do so, unless his holding be increased; but he keeps a donkey and a well-constructed cart. As yet he has no cow, not having his land in sufficiently clean order for laying down any part of it with grasses; but he has two yards full of pigs, which he keeps for the sake of the rich manure they supply. I do not advocate his system altogether, but merely relate the most striking features of it. His pig-yards are very commodious, and well arranged for weaning, fattening, &c.; and his stock now consists of a sow with ten young ones in one yard, and six store pigs in another. These are in fine condition—fed on vetches, rye (of which the grain is now, July 20, ripe), and wash, consisting of pollards and water; their food next week, and for some time after, will be beans, ripe and unripe, according to their

successive stages. These pigs are now ten months old, and have never been outside their yard, nor do they seem to be (compared with pigs of the same age which have had the run of the common) injured by confinement. Being always highly littered in the yard, having the sleeping chamber kept perfectly clean, and being abundantly fed, they sport about the straw, and seem quite contented. But without such care and comfort young swine will certainly not thrive in imprisonment.

Johnny will fatten up these pigs in October for sale in November, with barley-meal, pollards, toppings, and potatoes; and judging from his success last year under similar circumstances, they will weigh (at the age of fourteen months) nearly two cwt. each. He does not intend to sell any of his ten young ones until they shall have been fattened in the same way; but their mother will be put up as soon as possible after they shall be weaned. He does not expect to realize any ready money by rearing and fattening them; when sold, his stock will merely pay for their keep—he considers the large quantity of valuable manure a sufficient return.

He has hired a labourer to work with him, and will incur but little expense for horse-labour, as he and his assistant together are able to dig an acre very deeply in ten days; and he considers one such digging equal to three light ploughings; and from his experience of the last year, he is of opinion that spade-husbandry is far cheaper than that which is effected by the plough. As he reaps his vetches and rye for the pigs, he cuts out the stubbles with a bean-hoe for litter; and for the perfect cleansing of the ground before he digs it up, he collects the stubbles and clears them from earth with a little harrow drawn by the ass, and will pursue the same plan with all his stubbles. Last year he cut and bound half an acre of wheat himself with a fagging-hook, which I have described in my *Cyclopædia*, in one day; and he and his labourer intend to cut down an acre this year in the same way.

I could enumerate many other particulars of this man's excellent husbandry—such as burning the clay of headlands for manuring his turnip-crop and cabbage seedling beds—but I fear to be tedious, and therefore shall only add, that Johnny Halfacre is a true exemplification of the sacred proverb, that “the soul of the diligent shall be made fat.” He is always diligent (not only in seed-time and harvest, but all the year round), but never so busy with his field or garden crops as to choke the seed of God's word in his heart, and render *that* unfruitful by sloth or negligence. As far as I can judge, he does not permit his worldly to supersede his eternal interests; and as he knows the value of the *present* TIME, so does he estimate aright the infinitely superior importance of that which is *future*.

IDLENESS.—The worst vices springing from the worst principles—the excesses of the libertine, and the outrages of the plunderer—usually take their rise from early and unsubdued idleness.—*Farr's Discourses on Education*.

LIFE AND ITS ILLUSIONS.

“Lean not on Earth—’twill pierce thee to the heart—
A broken reed at best, but oft a spear,
On whose sharp point Peace bleeds, and Hope expires.”

YOUNG.

We are but Shadows! None of all those things,
Formless and vague, that flit upon the wings
Of wild Imagination round thy couch,
When Slumber seals thine eyes, is clothed with such
An unreality as Human Life,
Cherished and clung to as it is; the fear,
The thrilling hope, the agonizing strife,
Are not more unavailing there than here.
To him who reads what Nature would pourtray,
What speaks the night? A comment on the day.
Day dies—Night lives—and, as in dumb derision,
Mocks the past phantom with her own vain vision!

Man shuts the Volume of the Past for aye—
A blind slave to the all-absorbing Present,
He courts debasement, and from day to day
His wheel of toil revolves, revolves incessant;
And well may earth-directed zeal be blighted!
And well may Time laugh selfish hopes to scorn!
He lives in vain whose reckless years have slighted
The humbling truth which Penitence and grey
Hairs teach the Wise, that such cold hopes are born
Only to dupe and to be thus requited!
How many such there be!—in whom the thorn
Which Disappointment plants festers in vain,
Save as the instrument of sleepless pain—
Who bear about with them the burning feeling
And fire of that intolerable word
Which, inly searching, pierceth, like a sword,
The breast whose wounds thenceforward know no healing!

Behold the overteeming globe! Its millions
Bear mournful witness. Cycles, centuries roll,
That Man may madly forfeit Heaven's pavilions,
To hug his darling trammels:—Yet the soul,
The startled soul, upbounding from the mire
Of earthliness, and all alive with fears,
Unsmothered by the lethargy of years
Whose dates are blanks, at moments *will* inquire,
“And whither tends this wasting struggle? Hath
The living universe no loftier path
Than that we toil on ever? Must the eye
Of Hope but light a desert? Shall the high
Spirit of Enterprise be chilled and bowed
And grovel in darkness, reft of all its proud
Prerogatives? Alas! and must Man barter
The Eternal for the Perishing—but to be
The world's applauded and degraded martyr,

Unsouled, enthralled, and never to be free?”

Ancient of Days! First Cause! Adored! Unknown!
Who wert, and art, and art to come! The heart
Yearns, in its lucid moods, to Thee alone!
Thy name is Love; thy word is Truth; thou art
The fount of Happiness—the source of Glory—
Eternity is in thy hands, and Power—
Oh, from that sphere unrecognised by our
Slow souls, look down upon a world which, hoary
In Evil and in Error though it be,
Retains even yet some trace of that primeval
Beauty that bloomed upon its brow ere Evil
And Error wiled it from Thy Love and Thee!
Look down, and if, while human brows are brightening
In godless triumph, angel eyes be weeping,
Publish thy will in syllables of lightning
And sentences of thunder to the Sleeping!
Look down, and renovate the waning name
Of Goodness, and relume the waning light
Of Truth and Purity!—that all may aim
At one imperishable crown—the bright
Guerdon which they who by untired and holy
Exertion overcome the world, inherit—
The Self-denying, the Peaceable, the Lowly,
The truly Merciful, the Poor in spirit!

So shall the end of thine all-perfect plan
At length be realised in erring Man.

DONNYBROOK.

Verily, Donnybrook fair is, to all intents and purposes, “dead and gone;” for the modern wretched assemblage of hungry-looking cattle, dogs’-meat horses, measly swine, and forlorn-looking human creatures, obliged to content themselves with staring at the exterior of the show-booths, for want of the means to visit the interior, no more resembles the Donnybrook of the past, than a troop of the old “bulkies,” armed with their Arcadian crooks, and helmeted with their old woollen nightcaps, resembled a squadron of lancers.

Alas! alas! how every thing is altered! No longer does the quiet citizen dread the approach of Trinity Sunday; no longer does he think it necessary to barricade his windows, and postpone exterior painting for a week or two, in order to save his glass and the decorator’s labour from the nocturnal industry of the gentle College students.

The students never mustered in much force at Donnybrook, because it unluckily came during the long vacation; but there were enough at any time to kick up a shindy or scrimmage (by modern innovators called “a row”), for, between those who resided in town, and such as for various reasons kept the vacation within the College walls, a pretty decent muster could, upon an emergency, be called together.

It was upon the 26th of August—isn’t it strange that I should recollect the day of the month, though I forget the year!—that Bob O’Gorman, Dan Sweeny, Dick Hall, and a few other undergraduates of T.C.D., resolved to go to the fair and have a spree.

Dick was a little, delicate, effeminate-looking “ould crab,” and so smock-faced that he would easily pass for a girl, and a rather good-looking one, if dressed in female attire.

But Dick’s effeminacy was confined to his looks, for his muscular power far exceeded that of any man an inch or two more in stature, or a stone more in weight. He was a perfect master of the small-sword, had no match at single-stick; and woe to the unhappy wretch who fell under the discipline of his little bony fists, for he was an accomplished amateur in the science of pugilism, then but little known and less practised than subsequently by gentlemen.

On the present occasion it was resolved that Dick should sustain the character of a girl, and much fun was anticipated from the punishment that the remainder of the party would inflict upon any presumptuous individual who should dare to molest the modest fair one.

At the end of the double range of tents called “Dame-street,” was one called “the Larkers;” and as this was uniformly crowded by citizens of Dublin, it was scarcely possible for any one, residing but for a month in town, not to be recognised by some person present, who immediately passed the name of the new-comer round, and he was surprised (if a raw one) to hear himself addressed by name, by persons whom he never saw in his life before.

It was at the entrance of this tent that a countryman stood, attired in the usual large frieze over-coat (which, from its being worn in summer as well as winter, might lead a stranger to suppose that there seldom or never is a hot day in Ireland), and accompanied by a pretty, bashful-looking girl,

apparently fresh from the “interior.” After gaping for a considerable time, some gentlemen, amused by the wonderment that he exhibited, and probably somewhat touched by his companion’s charms, called to him to “come in.” With some reluctance he accepted the invitation, and, fearful of intruding upon the “gintlemin,” seated himself awkwardly upon the end of a form; up *it* tilted, and down *he* went, to the great delight of the beholders. Having gathered himself up, he reseated himself more firmly, placing “Biddy” near him, she having declined all offers of other accommodation pressed on her by the company.

Paddy O’Neill (the name by which he announced himself), having been pretty well plied with punch, had grown very voluble, and seemed to be beginning to feel himself quite at home, had told many queer stories, and made his entertainers laugh very heartily, when two elderly gentlemen, closely muffled, entered rather stealthily, and sliding over, suddenly seated themselves behind Paddy. Biddy, who had been hitherto quite silent, answering every compliment or remark addressed to her only with a smile, gave Paddy a nudge, and whispered something into his ear, that caused him to turn and gaze at the new arrivals.

“Arrah, thin, Docthor M——, agra, who’d ha’ thought o’ meetin’ *you* here?” said he, addressing one of them, who sprang at the mention of his name, as if he had sat on the point of a stray nail; he and his companion Dr H——, both senior fellows of Trinity College, having disguised themselves, as they thought effectually, for the purpose of seeing, for the first time in their lives, the fair, and the fun of it, without being recognised in such an uncanonical assemblage. With this object they had avoided exposing themselves to the risk of walking down the tent, but had merely slipped in to reconnoitre from behind the shelter of the frieze-coated customer, who now, so inopportunistly and innocently, had announced the name of one of them.

“Hold your tongue, sir!” said Dr M.; “you mistake me, sir.”

“Arrah, docthor darlint, sure iv I mistake ye, ye need’nt get into sich a comflusthration about id; bud sure I know ye too well to mistake ye. Sure, aint I the boy that had the misforthin to dhrop yer honor’s riverince into the bog-hole, whin ye wint out to make believe ye were snipe shootin’, down at Colonel Thrench’s, last Candlemas was a twelmonth.”

“I don’t know you, sir!” roared the doctor in an agony, hoping by his ferocity to overawe the countryman into silence; but Paddy had taken too much punch to notice the tone, and seemed incapable of entertaining or following up more than one idea at a time, and the one now before him was that of forcing himself, will he nill he, upon the recollection of the worthy doctor.

“Ye don’t know me!—well, listen to that!—ye don’t know me!—oh, well, iv that does’nt flog! Arrah, thin, maybe ye don’t recollect the bog-hole that ye wanted me to carry ye over, an’ ye war so mortal heavy that my fut slipped, an’ I had the luck to fall an my face, jist at the very edge iv the slush, an’ ye pitched right-over, head foremost, into the very middle iv id; an’ iv id was’nt for the good luck that yer legs stuck out, jist the laste taste in life, by which I got a hould iv ye, sure would’nt ye be lost intirely? An’ don’t ye”——

“Hold your tongue, you infernal scoundrel!” roared the enraged doctor, who saw that every eye was fixed upon him, and every one’s attention drawn to the spot, from the eagerness of manner and stentorian voice of Paddy, whose reminiscence had produced a roar of laughter. Escape, too, was utterly hopeless, for the tent had been filling, and the doorway was blocked up by those who were pressing forward from the outside to get a view of the speaker. “Hold your tongue, sirrah; you mistake me for some one else. I never was thrown into a bog-hole in my life.”

“Oh! pillelieu! meellia murther! listen to that—as iv any one that iver seen Docthor M— ov Thrinity College could iver mistake him agin; bud sure Docthor H— there ’ill may be help out yer

mimory [Dr H— gave a writhe, for he had hoped to have escaped, at least]; sure he was at the colonel's whin ye war brought home in the muck."

This announcement of the names and address of both the unfortunate betrayed, was received with a shout, whilst Paddy's earnestness to free himself from the charge of having blundered, increased every moment, and reminiscence followed reminiscence, each in a louder tone than the preceding, until his argument became a perfect shout, whilst the unlucky S.F.T.C.D.'s strove to out-bellow him with their denials, and the audience laughed, shouted, and danced with glee at the fun.

"I protest," bawled Dr H—, "that I do not know Colonel Trench. You mistake, my honest man; I never was at his place in my life. My friend here, Dr M—, knows him, and has been there often; but I have not, I assure you."

"Oh! you ass," bellowed Dr M—, "what do you acknowledge my name for? 'Tis no wonder they call you 'Leatherhead H—.'"

A renewed roar followed this piece of blundering recrimination.

"Never at Colonel Thrench's!—not you!—oh! ye desavin' ould villain!" screamed the hitherto silent Biddy. "Not you!—Do ye know me!—do ye!—do ye!!—Do-o-o-o-o ye!!!" every repetition of "do ye" being louder and longer than the last, until she finished in a terrific long shriek, squeezing her hands together upon her knees, and stamping alternately with her feet, with a rapidity that gave the effect of a shake to her voice.

"I do protest and declare," shouted the worthy doctor, "that I never, to my knowledge, saw your face before."

"Arrah, Biddy, avourneen, is this the ould Turk that ye tould me about, bud would'nt mintion his name, that was so imperant to ye? Scraub his face, the ould thief! and let me see iv he dar purvint ye, my darlin'. Tache him to behave himself to unpurcted faymales!"

Biddy, who seemed quite inclined to forestall her companion's orders, had sprung upon the unlucky doctor before the sentence was half finished. He strove in vain to shake her off; she clung to him like a wild-cat, screaming, shrieking, scolding, biting, scratching, and tearing, until at length she maddened him past all endurance by pulling two handfuls of hair successively out of the little that remained on his skull, for which he repaid her with two furious blows.

The spectators, who had hitherto looked on, and merely laughed at the entire affair as an excellent joke, had undergone a change of sentiment upon hearing the innuendo contained in Paddy's last speech; and, no longer considering the old gentlemen as a pair of innocents amusingly "blown," they now looked upon them as a pair of wicked old profligates, worse than young ones; and one, more zealous than the rest, shouting out "shame! to strike the girl," stretched Dr H— with a blow.

Dr M., irascible at all times, now lost all self-possession, and, unable to reach his friend's new assailant, turned furiously upon the cause of all his woe, and bestowed a shower of blows with his stick upon Paddy, before the latter had time to bring his cudgel to parry them. He soon recovered himself, however, and from defendant quickly became assailant.

Many of the bystanders indignantly called out, "Murder the ould villain—knock out his brains, Paddy. That's right, Biddy; flitther him!" and several proceeded to give a helping hand to the good work; but others thought it was a shame for a whole lot of people to fall upon two, and in their love for justice they ranged themselves alongside the reverend doctors, shouting, "fair play's a jewel!" The fight thickened, volunteers joining either rank every moment, in the laudable endeavour to keep up the balance of power. Biddy had quitted her grip of the doctor, and was now,

to the surprise of those who had time to look about them (and they were few), engaged in the endeavour to wrench a stick out of the hands of a huge hulk of an Englishman, who, having merely gone to see the fun at Donnybrook, without the most remote idea of joining in a fight, could not be persuaded of the necessity of giving his stick, as he did not intend to use it himself, to one who *did*, and that one “a female!” At first he laughed; but he was quickly obliged to put forth all his strength to retain it, and, whilst twisting about, he caught a stray blow that floored him; he fell against a table, which of course overset; the confusion increased, when a shout suddenly arose, “Hurrah for Dr M—! Hurrah for Dr H—! College to the rescue!—Trinity!—Trinity!”

At the well-known war-cry of the students, several changed sides; those who had just been defending the doctors now turned upon them, whilst many of their late assailants ranged themselves on their side. The citizens, thinking that the number of students must be small, rushed to the spot, to pay off sundry old scores; but one would imagine that the cry of “Trinity! Trinity!” which resounded on all sides, was a sort of spell, or incantation, that raised spirits from the earth, so many voices responded to the call.

The unfortunate doctors, who had just expected nothing short of utter annihilation, felt their spirits rise at the prospect of aid and rescue, and bellowed with might and main, “Trinity! Trinity!” and in a few minutes they were the nucleus of a fight in which the whole fair had joined.

“The poliss!—the poliss!—here come the bloody poliss!” was now the cry; and the horse police dashed into the mob with their customary ardour, their spurs fastened in their horses’ flanks causing them to plunge, and bite, and kick most furiously, and laying about them with their swords, cutting at every thing and every one within their reach; luckily they did not know the sword exercise, and, therefore, when they struck with the edge, it was only by accident. In a jiffy, the reverend seniors, caught in the very act of shouting “Trinity!” were handcuffed, as were also the Englishman, who got a blow of a sabre from a policeman that nearly took off his ear, for attempting to expostulate; Paddy, who submitted quietly; and Biddy, after a severe tussle, in which she reefed one policeman’s face, and nearly bit the thumb off another. They were all put together into a jingle, and conducted by a mounted escort to town; the police hurrying them for fear of a rescue, by keeping continually whaling the driver with the flats of their swords, and prodding the horse with the points, which so enraged the jarvy, that when he got near the corner of Leeson-street, Stephen’s-green, where two or three hundred of his brethren were assembled, having whipped his Rosinante into a gallop, he drove against a brewer’s dray, by which his traces were smashed, his horse set free, the jingle locked fast, and he, springing off his perch, shouted out, “down with the bloody poliss!”

In an instant the mob rushed upon them. Paddy and Biddy, with an alacrity and agility truly astonishing, sprang from the lofty vehicle, plunged into the crowd (where there were plenty of willing hands to free them from the handcuffs), and escaped. Nor were the worthy doctors slow in following their example, the only prisoner that remained being the bewildered Englishman, who suffered “only” a three months’ incarceration in his majesty’s jail of Newgate for going to see Donnybrook, and the fun at it, his sentence having been mercifully mitigated, in consideration of its being his first offence!

“Well,” said Dr H—, when he went with his head bandaged up, a shade over his right eye, and about twenty bits of sticking plaster stuck over his face, to visit Dr M— (who was unable to leave his bed for a week), “well, what a fool I was to be persuaded by you to go to Donnybrook fair! what a pretty exhibition we would have made at the police office this morning! Was it not most fortunate that we made our escape?”

“I have been thinking,” said (or rather groaned) Dr M—, “who that scoundrelly country fellow could be. I never fell into a bog in my life—that was all a lie; and still the blackguard’s face was familiar to me.”

“I think he was very like that scapegrace Robert O’Gorman, only that he had light hair; and though I could take my oath I know nothing of that infamous little wretch that they called Bidy, yet I do think I have seen her face before—hum”—

“Could it have been that he disguised himself, eh! I’ll inquire into it, and if he did, by”—

“I think, my dear M—, you had better let it alone; the less we say about it the better. You know we really led the fight—that’s a fact that can’t be denied; though it surprises me how we were hooked into it.”

A rustle at the door, followed by a loud knock, announced that the newspaper had been thrust into the letter-box, from which Dr H— immediately extracted it; and as he glanced over the page, the following paragraph met his eye. It was headed “Disgraceful and fatal riot at Donnybrook:”

“It is with mingled feelings of indignation, horror, and contempt, that we feel bound, in discharge of our imperative, onerous, and painful duty to the public, to give publicity to one of the most astounding, frightful, and overwhelming facts which it has ever fallen to our lot, as faithful journalists, to record. The peaceable, gentle, and innoxious inhabitants of the village of Donnybrook, and the casual visitors who sought a little innocent recreation at the fair now being holden, were yesterday evening thrown into a state of the utmost alarm, confusion, and dismay, by a barefaced attempt to carry off by brutal force a young girl from the guardianship and protection of her brother. It appears that they had gone into a tent to rest and refresh themselves (having probably over-exerted their light fantastic toes), when their savage assailants (respecting whose rank and station various rumours are afloat, which for the present we forbear from mentioning) rushed upon them, and endeavoured to force her away. The indignant bystanders interfered to prevent the outrage, when—will it, can it be believed? our pen trembles, and a cold thrill runs through us as we write it!—the worse than Indian war-whoop, the yell of the collegians, was raised, and their numbers would in all human probability have succeeded, but for the timely interference of the police, to whose humanity, promptitude, and forbearance, upon the trying occasion, too much praise cannot be given. The riot was not quelled until the military were called out, and by three o’clock this morning all was again quiet. Up to the time of going to press we had only heard of sixteen lives being lost.

Second Edition.—We stop the press to announce that no lives have been lost; but Sir Patrick Dunn’s, the Meath, and Mercer’s hospitals, are crowded with wounded. N.B. The soldiers were not called out.

Third Edition.—Dr Fitzgerald has just informed us that there are no wounded in either Sir Patrick’s, the Meath, or Mercer’s.”

“Well,” said Dr H—, “if they are not there, we at least know where some of them are.”

NAISI.

WHAT IS THE USE OF WATER?

Why is it that of the whole surface of this globe, we may consider that three-fourths are covered by water, and that only one-fourth is in a condition to be permanently inhabited by human beings? Is there any great object in nature served by this? Is there any law of nature which would prevent the proportion being one-fourth water to three-fourths land, or even less water? In fact, what after all is the great use of water upon the large scale in nature?

First of all, although three-fourths of the globe are now covered with water, there is no reason to suppose that it has been always so. On the contrary, it is quite certain that the proportion between land and water has changed very much and very frequently; that the whole continent of Europe was at one time the bed of an immense sea, when probably there was a great continent where the Pacific Ocean is now spread; that even Old Ireland was once not merely what Admiral Yorke wished her to be, forty-eight hours under water, but probably many thousand years in that condition; and that the great tract of limestone which occupies all the centre of the country, is nothing more than a collection of the skeletons of shell-fish, her first inhabitants, which by time and pressure have been converted into the hard material of which we build our houses, and which we burn into lime. There is thus no particular reason why there should be three times as much water at present as land, but it is easy to show that water on the great, as well as on the small scale, is of paramount importance in nature.

Water is a portion of the food of all living beings. In the case of animals, the bodies from whence they derive nutriment are so varied and so complex, that to illustrate the peculiar part which water plays in each, would occupy too much space. In all our drinks, even in ardent spirits, there is a very large quantity of water, and our solid food very seldom contains less than nine-tenths of its weight of water. The living body is even less solid. A man weighing 150 lbs. would, if perfectly dried, weigh not more than 10 lbs., the other 140 lbs. being water. It is to the existence of this quantity of water that we owe the elasticity, the softness, and pliability of the different portions of our frame, the animal tissues being, when dry, hard and brittle as dry glue.

The nutrition of vegetables furnishes a beautiful and simple example of the use of water in nature. The body of the vegetable, the proper wood, may be considered as being composed of water and of charcoal; and hence, when we heat a piece of wood until we decompose it, the water is expelled, and carbon or charcoal remains behind. In order to grow, a plant must therefore get water and charcoal in a form fit for its use, that is, in such a form as it can make food of, and digest them. For this, the carbon is supplied in the carbonic acid which the air contains, and the water in the state of vapour which the air contains also, and which is continually descending under the form of dew and rain to moisten the leaves and the roots of the plants, when it has been absorbed into the ground. All the water which is absorbed by plants is not assimilated, or digested; a great part is again thrown out by the surface of the leaves; for, precisely as the air which an animal expires from the lungs in breathing is loaded with vapour, so is there a process of perspiration from the surface of the leaves, which are the lungs of plants. For the formation of substances which are peculiar to certain plants, other substances are required as food, thus, most plants require nitrogen, which is

accordingly furnished abundantly in atmospheric air; others must have access to sulphur, in order to flourish; but this depends, as it were, upon particular branches of manufacture in which the plant is engaged; for its own support, for making wood, and the tissue of its leaves and vessels, it uses only water and carbonic acid.

The conversion of water into steam or invisible vapour by boiling, is one of the best known facts in science; but by a little attention we can observe that this change takes place at almost all temperatures, although much less rapidly. Thus, if a little water be laid in a plate, it is soon dried up, and wet clothes, by being hung up in the air, are very soon completely dried. Even below the temperature at which water freezes, it still evaporates; and thus, when a fall of snow is succeeded by a continued frost, the snow gradually disappears from the fields without having melted, evaporating while yet solid. From the surface of all the water of the globe, therefore, there is continually ascending a stream of watery vapour; but as the proportion of sea is so much greater than that of land, we may look upon the ocean as being the source of the watery vapour of the air upon the large scale.

Now, watery vapour is lighter than air, and hence the vapour, as soon as formed, ascends in the air like a balloon, until it arrives at a part of the air which is of its own specific gravity. The air in these higher regions is extremely cold, and the vapour can no longer maintain itself under the form of invisible steam: it is condensed, and would immediately fall back to its source as rain or hail, but for a singular property which it acquires at the moment of being vaporised. When water evaporates, it becomes highly electrified, and could attract a feather, or other light bodies, like a stick of sealing-wax which has been rubbed briskly on a woollen cloth. Now, the vapour which passes off is electrified also; and while in this state of electricity, it, on arriving at the colder regions of the air, cannot condense, to form liquid water. The minute particles of the water repel each other too violently, in virtue of their electricities, to form drops, but they constitute the great loose collections of clouds which diversify so much the appearance of our sky. The clouds being thus highly electrical, and being very light, are attracted by the tops of mountains and high lands, or by elevated buildings; and, giving off their electricity, the particles of water coalesce, to form drops which descend as rain. In this country the air is so damp that in general the discharge of the electricity of the clouds takes place quietly and silently; but in summer, and in dry climates, it produces the vivid flashings and injurious effects of the lightning, and the re-echoed rattle of the thunder-clap.

When water is cooled, it diminishes in bulk like other bodies; but at a particular temperature it deviates from the general law of contraction, and by doing so, becomes, perhaps, the most striking example of providential design that is to be met with in inorganic nature. Cold water is specifically heavier than warm water, in consequence of the contraction it has undergone, and hence will sink in it, as water would sink in oil. Now, if we consider the surface of a lake exposed to the cooling action of a wintry wind, the water which is first cooled becomes heavier, and, sinking to the bottom, is replaced by the warmer water, which floats up to the top; there is thus a current established of cold water descending and of warmer water rising up. This continues until all the water in the lake has been cooled down to the temperature at which its specific gravity is greatest, which is about 40 degrees, or about eight degrees above the point at which it begins to freeze. The action of the cold wind continuing, the water at the surface is still further cooled; but now, in place of contracting, it expands—instead of becoming heavier, it becomes lighter, and remains floating upon the surface. It is then still further cooled, and finally its temperature being reduced to 32 degrees, it freezes, and a layer of ice is formed on the surface of the lake. This ice, and the cold water next it, are impermeable to heat: it actually serves as a blanket to the water at 40 degrees which is below, preventing the escape of the heat, and retaining it at that temperature, sufficient for

the purposes to which it is subservient; for at the temperature of 40 degrees, the life and enjoyments of all the various tribes of animals and vegetables which reside permanently under the surface of the water are perfectly secured, at least for a very considerable time; the water holding dissolved a quantity of oxygen for the animal respiration, and the vegetables living on the carbonic acid which is formed by the respiration of the fish. On the approach of spring, the warmer air, and the rays of the more elevated sun, act directly on the surface of the ice, and each portion of water formed by melting, becoming heavier, sinks, so as to expose the ice itself to the source of heat. Thus the ice is rapidly dissolved, and after a few days the lake throws off its wintry aspect altogether.

Now, if water did not possess this peculiarity of being heaviest at the particular temperature of 40 degrees—if it contracted according as it was cooled, up to the moment of freezing, as almost all other liquids do, what would be the result? The cold wind acting on the surface of the lake, and the water becoming heavier by being cooled, the circulation would continue until all the water had been cooled to the point at which it freezes. The ice would then form indifferently in all portions of it, at the bottom and in the centre, as well as on the surface; and by the continued action of the source of cold, the wind, the whole mass of water in the lake would be frozen into a solid block of ice. The watery sap in the vessels of the aquatic plants, the blood in fishes and other animals inhabiting the water, would be equally frozen, and all these living beings consequently killed. Further, on the approach of summer, by the first heating action of the air and sun, a layer of ice, of a few inches thick upon the surface, would be melted, but the water thus produced would, by being impenetrable to heat, prevent the great body of ice below from being affected. Just as, in reality, the cold water at the surface prevents the warmer water below from being cooled, so then it would prevent the colder ice below from being warmed; and hence the heats of summer passing over without the melting process extending beyond a few feet in depth, the first cold days of the next winter would solidify all again.

In every country, therefore, where at present water is frozen at all in winter, we should have there established the reign of perpetual frost. By the presence of such large masses of ice, the temperature of the ground would be so much reduced, that, in place of the rich herbage of our meadows, and the luxuriant produce of our corn-fields, we should have our country yielding a scanty support to wandering herds of deer, in the mosses and lichens that could be scraped up from beneath the snow. The oaks, the beeches, the horse-chesnuts, which give such beauty to our sylvan scenery, would disappear, and the monotony of wildernesses of the Scotch fir and of the spruce would be varied only by patches of stunted birch. The countries nearer the tropics would be gradually brought into the same condition, by the depression of their mean temperature; and thus, in a short time after water had ceased to possess this peculiar property, the whole surface of the globe would be reduced to the condition of which we now happily only read in the tales of the arctic voyagers; and all commerce, manufactures, and civilization, would be banished from the earth. Of such value is this little peculiarity of water!

A property of water, which, however, unlike the former, it shares with all other liquids, is, that when it freezes it gives out a large quantity of heat; and that conversely, in order that ice may melt, it must obtain, from some other source, a quantity equally considerable. Consequently, water freezes and ice melts very slowly; and that it should melt thus slowly, is of essential importance in animated nature. If in spring or summer, when vegetable life is in activity, when the development of leaves, of flowers, and fruit, is at its greatest energy, and all the vessels of the plant are distended with its nutritious juices, were it suddenly exposed to cold, the sap would be frozen, and by the expansion of the ice the vegetable tissues torn to pieces, and the plant killed. In the thin extremities, as in the leaves, such is the effect of the frost of a single night; but as the fluids,

yielding but gradually up their latent heat, solidify very slowly, the injury does not extend so far as to be beyond the remedial powers of the plant itself. In another way, however, the peculiar latent heat of water is of still more importance. If there was no large collection of water on the globe, the change of seasons would be amazingly more rapid and more remarkable than they at present are. A change in the direction of the wind, the alteration which a few weeks should effect in the position of the sun, would transfer us from the depth of the severest colds of winter to the summer heats. These colds and heats would also be much greater than they at present are, and an approximation to this actually occurs in countries far distant from the sea. The central districts of Europe and of Asia have what are termed continental climates to distinguish them from ours, which is called insular. Their summers are hotter, their winters are much colder, and the spring and autumn seasons of passage, which with us might be said to occupy most of the year, are in those countries of only a few weeks', or even a few days', duration. In fact, when on the cessation of summer the first cold winds tend to bring on the winter, and to bind up our lakes in frost, the first portion of water frozen becomes, by giving up its latent heat, a source of warmth which tempers the chilly air, and retards its action on the remainder. The water freezes thus very slowly. The vegetables, and certain classes of animals, feeling the cold of winter thus gradually coming on, prepare to meet it without injury. The motion of the sap in the one, that of the blood in the other class of living beings, becomes slower, and, dropping its leaves and fruit, the tree retains but its firm trunk, within which its energies are preserved for the ensuing season; whilst the hedgehog, the viper, the frog, and other animals, retire to their hiding-places, and in a state of almost lifeless stupor remain until the warmth of the succeeding spring calls them to renewed existence.

In the formation of the insular climate which we possess, another power of water, however, equally or perhaps more influential, can be traced. There issues continually from the ocean at the equator, as the earth revolves, a current of water considerably warmer than that which bathes our shores. This current becoming sensible first in the Gulf of Mexico, is called the Gulf Stream; it passes obliquely across the Atlantic, floating on the colder water of the ocean, which tends in a direction nearly opposite to replace it, and thus diffuses over the coasts of North America and Europe the heat which it had absorbed within the torrid zone. The northerly winds, which would bring down a sudden winter on us, are therefore tempered by passing over the warmer surface of the ocean; whilst the hot winds from the south, which on the approach of spring might make too premature a change, expend, in passing over the great expanse of sea, a portion of their heat; and thus the transition in both directions is rendered more gradual and harmless.

These are but a few of the important duties which are allotted to water in its place in nature. It in other respects presents an equally interesting subject of examination, and it is one to which we shall return. From its value as the great agent of nutrition to the vegetable world, and the necessity of a supply of it to animals; from its power in modifying the appearance and structure of a country, changing land into sea, and elevating banks where deep water had been before, the philosophers of old looked upon water as the origin of all earthly things, as being above all others the element of nature. It is not so: water is not an element. Among other wonders which chemistry has taught us, we have learned of what water is composed; and on another occasion we shall describe the way in which its elements may be obtained.

K.

*** END OF THE PROJECT GUTENBERG EBOOK THE IRISH
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